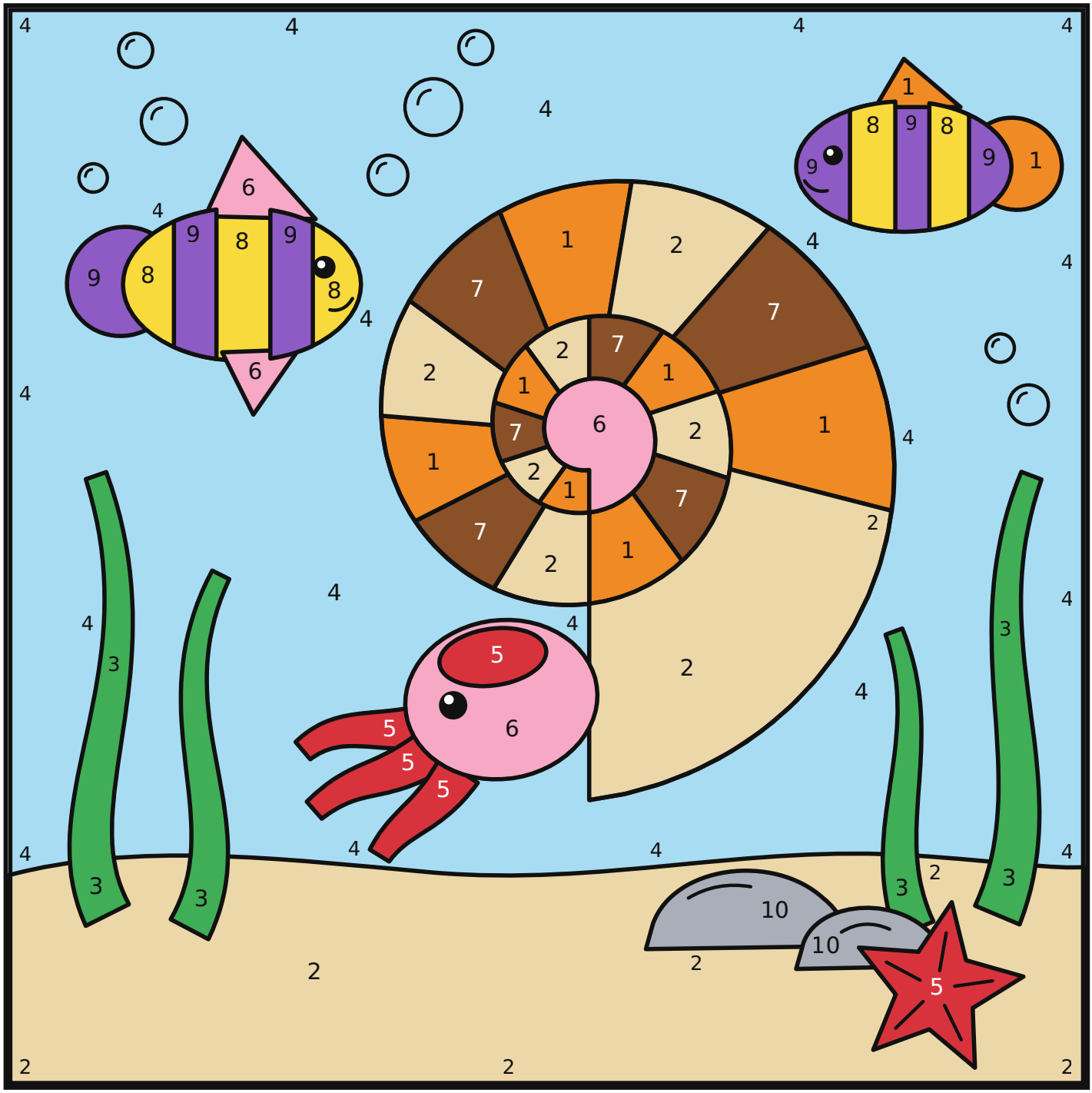
Name: **ANSWER KEY** _____ Date: _____

# Exponential & Logarithmic Functions: Color By Number



**Color Code:**

| | | | | |
|---|---|---|---|---|
| **9** Purple<br>x = 16 | **1** Orange<br>x = 7 | **5** Red<br>3 | **10** Gray<br>y = −5 | **4** Light Blue<br>x ≈ 1.65 |
| **6** Pink<br>x = 1 or x = 2 | **2** Tan<br>$\frac{5}{2}$ | **8** Yellow<br>t ≈ 46.2 | **7** Brown<br>$2,440.38 | **3** Green<br>x = 5 |

Name: **ANSWER KEY** _____ Date: _____

# Exponential & Logarithmic Functions: Color By Number

**Directions:** Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch closely: the answers look alike!

**1.**

Solve: $8^{x-1} = 4^{x+2}$

Write both sides with base 2: $2^{3(x-1)} = 2^{2(x+2)}$

$3x - 3 = 2x + 4$

**x = 7**

1. Color: **Orange**

**2.**

Evaluate without a calculator: $\log_4 32$

Let $\log_4 32 = y$, so $4^y = 32$

$2^{2y} = 2^5$, so $2y = 5$

$\frac{5}{2}$

2. Color: **Tan**

**3.**

Solve: $\log_2(x+3) + \log_2(x-3) = 4$. Check for extraneous solutions.

$\log_2[(x+3)(x-3)] = 4$, so $x^2 - 9 = 2^4 = 16$

$x^2 = 25$, so $x = 5$ or $x = -5$

$x = -5$ makes $x + 3$ negative: reject it

**x = 5**

3. Color: **Green**

**4.**

Solve: $5^{2x-1} = 40$. Round to the nearest hundredth.

$(2x-1)\log 5 = \log 40$, so $2x - 1 = \frac{\log 40}{\log 5} \approx 2.2920$

$x = \frac{2.2920 + 1}{2}$ (round only at the end)

**x ≈ 1.65**

4. Color: **Light Blue**

**5.**

Evaluate: $3\log_5 10 - \log_5 8$

Power rule: $\log_5 10^3 - \log_5 8 = \log_5 1000 - \log_5 8$

Quotient rule: $\log_5 \frac{1000}{8} = \log_5 125$, and $5^3 = 125$

**3**

5. Color: **Red**

**6.**

Solve: $4^x - 6 \cdot 2^x + 8 = 0$

$4^x = (2^x)^2$. Let $u = 2^x$: $u^2 - 6u + 8 = 0$

$(u-2)(u-4) = 0$, so $2^x = 2$ or $2^x = 4$

**x = 1 or x = 2**

6. Color: **Pink**

**7.**

$2,000 is invested at 4% annual interest compounded quarterly. Find the balance after 5 years, to the nearest cent.

$A = P(1 + \frac{r}{n})^{nt} = 2000(1 + \frac{0.04}{4})^{20} = 2000(1.01)^{20}$

**$2,440.38**

7. Color: **Brown**

**8.**

A 500 mg sample decays by $A = 500e^{-0.03t}$, (t in days). After how many days will 125 mg remain? Round to the nearest tenth.

$0.25 = e^{-0.03t}$, so $\ln 0.25 = -0.03t$ and $t = \frac{\ln 0.25}{-0.03}$

**t ≈ 46.2**

8. Color: **Yellow**

**9.**

Solve for x: $\log_x 64 = \frac{3}{2}$

$x^{\frac{3}{2}} = 64$, so $x = 64^{\frac{2}{3}}$: take the cube root first (4), then square it

**x = 16**

9. Color: **Purple**

**10.**

Find the horizontal asymptote of $f(x) = -3 \cdot 2^{x+1} - 5$

As $x \to -\infty$, $2^{x+1} \to 0$, so $f(x) \to -3(0) - 5$

The shifts inside the exponent do not move the asymptote.

**y = −5**

10. Color: **Gray**